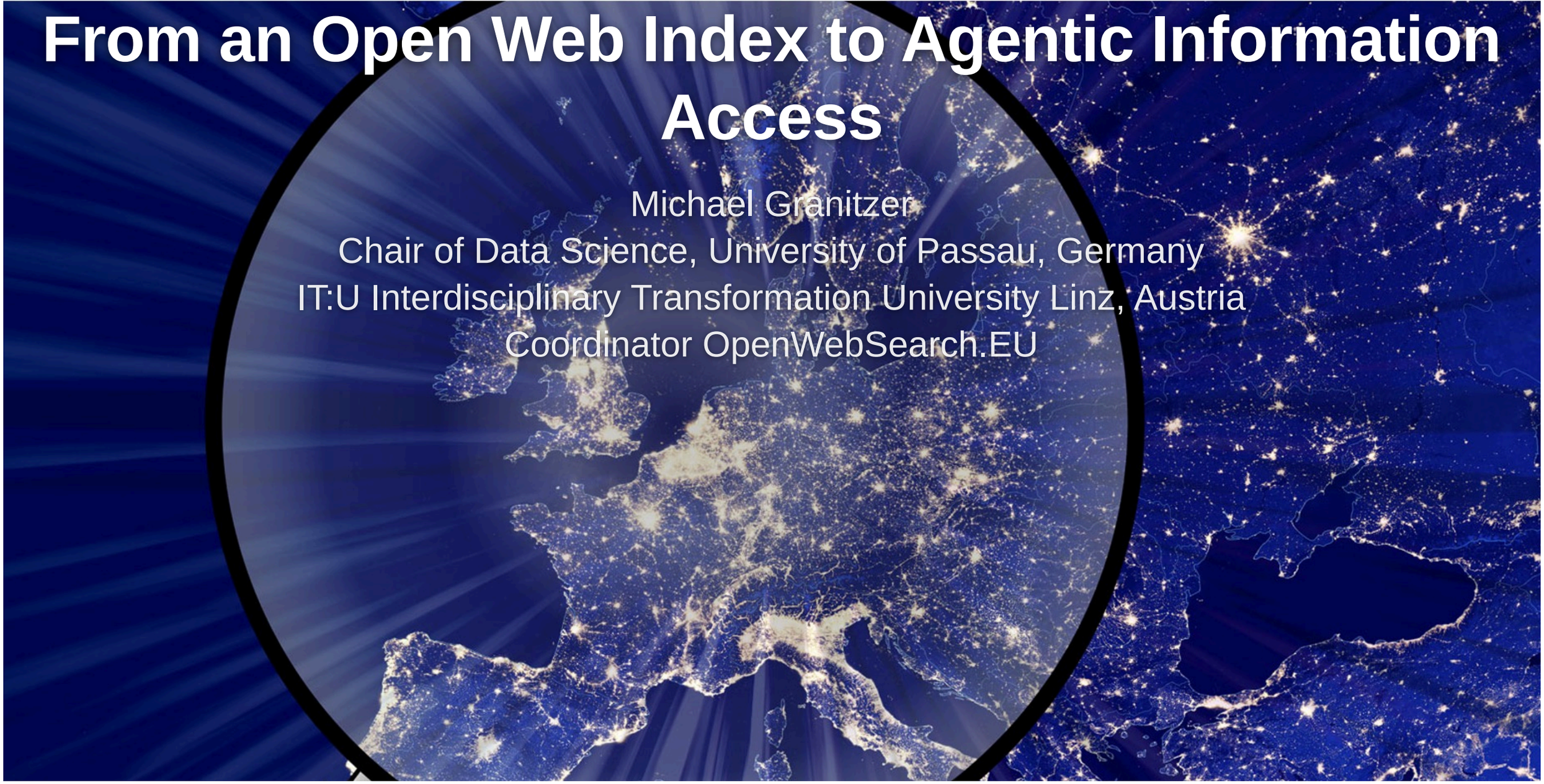
From an Open Web Index to Agentic Information Access

Michael Granitzer
Chair of Data Science, University of Passau, Germany
IT:U Interdisciplinary Transformation University Linz, Austria
Coordinator OpenWebSearch.EU



Funded by

Bundesministerium für Forschung, Technologie und Raumfahrt

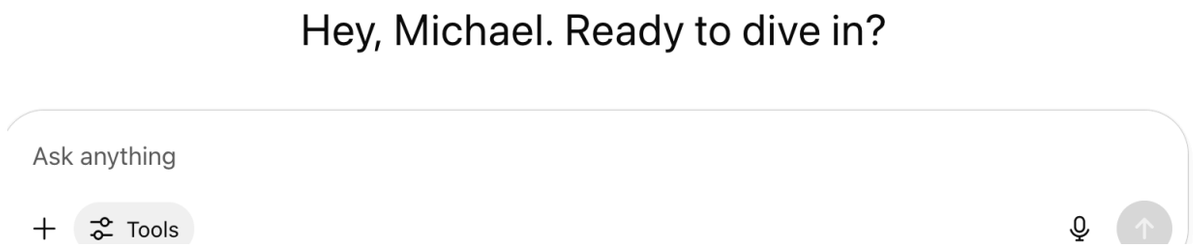
Bundesministerium für Landwirtschaft, Ernährung und Heimat

Bayerisches Staatsministerium für Wirtschaft, Landesentwicklung und Energie

1 · Is search dead?

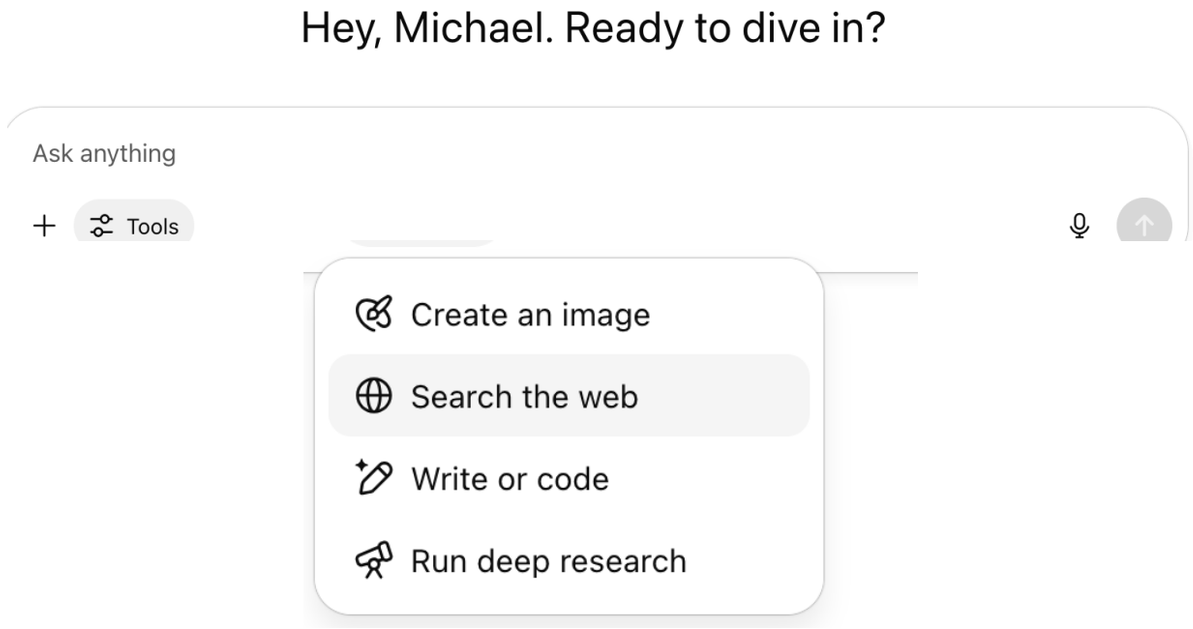
AND WHAT WILL HAPPEN WITH THE 10 BLUE LINKS?

Search is dead



Semrush clickstream analysis, Apr 2026 · Chatterji et al., How People Use ChatGPT, NBER 2025 · Chen et al., BrowseComp-Plus, 2025 (GPT-5, o3).

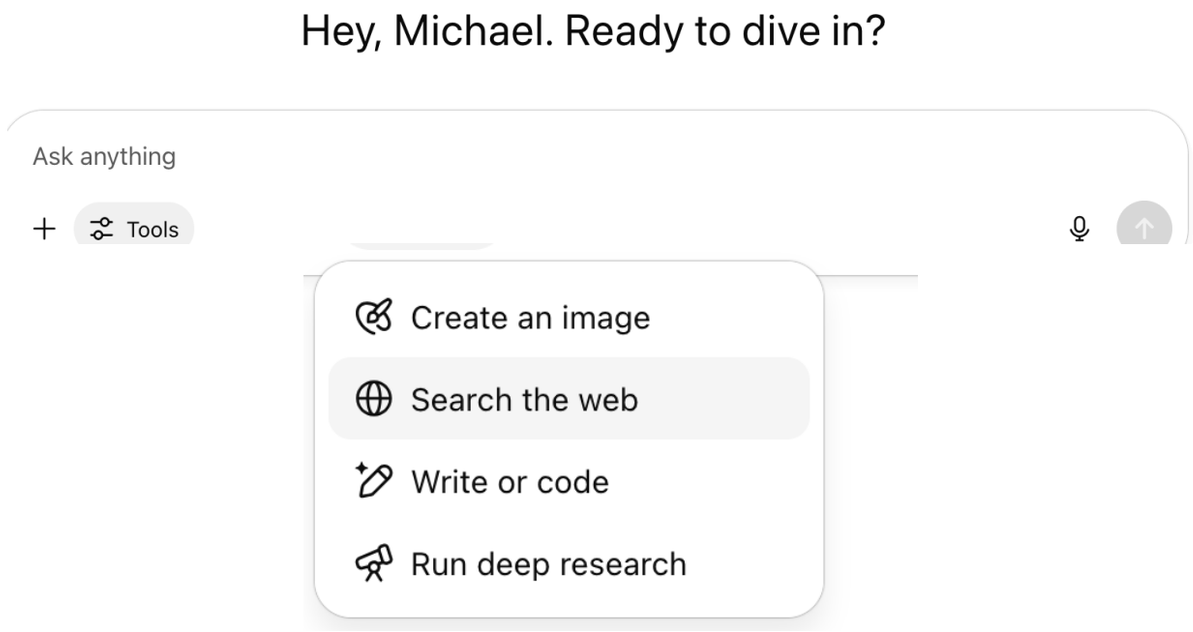
Search is dead



"Search the web" sits behind the box, switched on by default in most assistants.

Semrush clickstream analysis, Apr 2026 · Chatterji et al., How People Use ChatGPT, NBER 2025 · Chen et al., BrowseComp-Plus, 2025 (GPT-5, o3).

Search is dead — long live search



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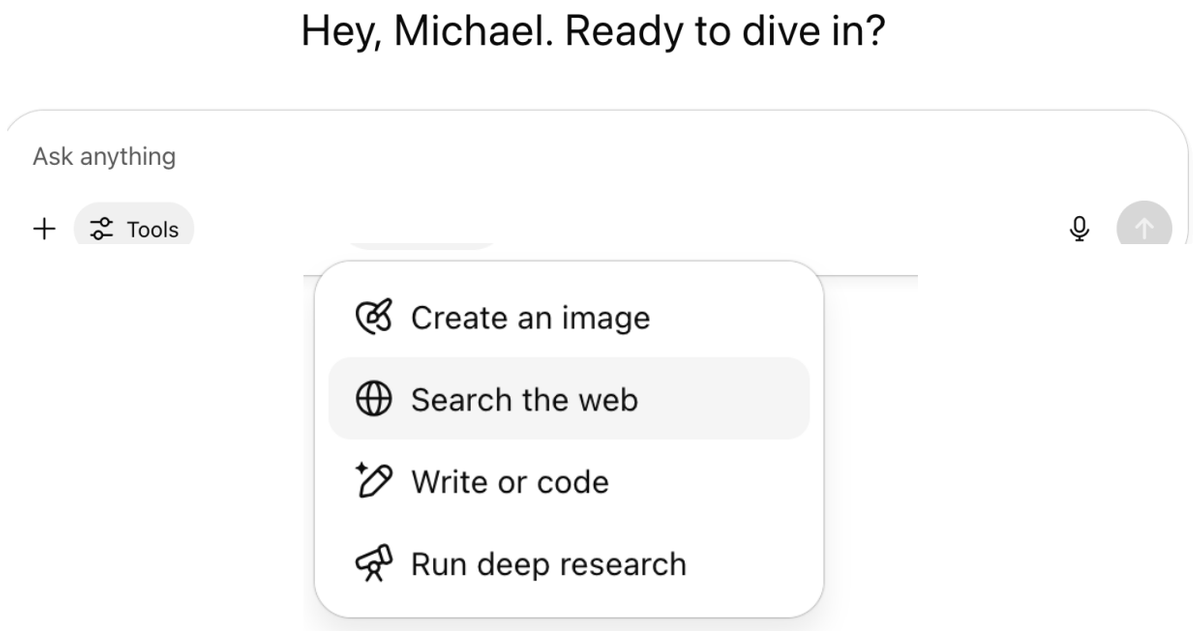
34.5%
of ChatGPT queries switched on web search in February 2026

14% → 24%
of ChatGPT usage is "seeking information" in one year, "a very close substitute for web search"

20–26
search calls per question by a deep-research agent

Semrush clickstream analysis, Apr 2026 · Chatterji et al., How People Use ChatGPT, NBER 2025 · Chen et al., BrowseComp-Plus, 2025 (GPT-5, o3).

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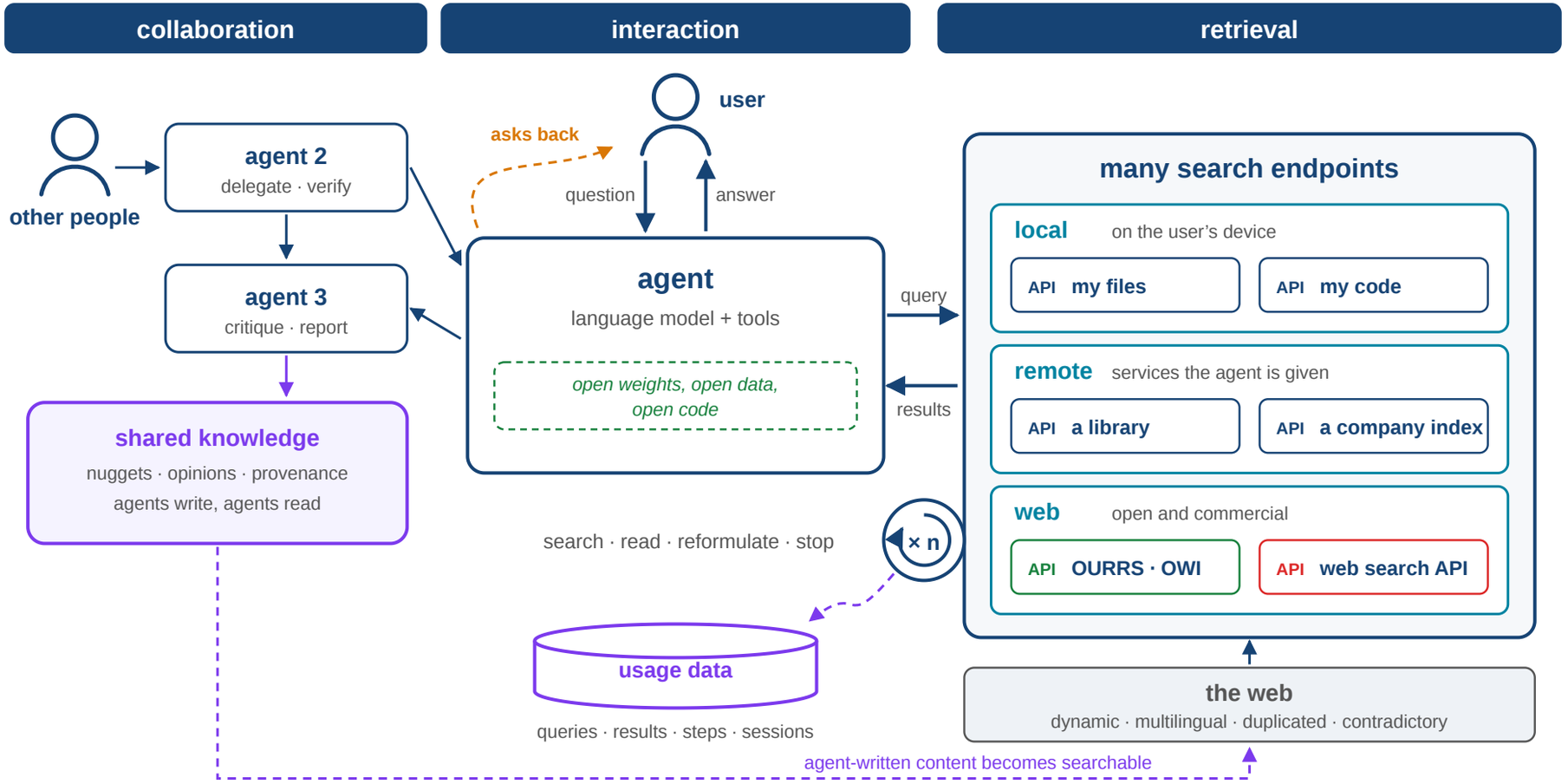
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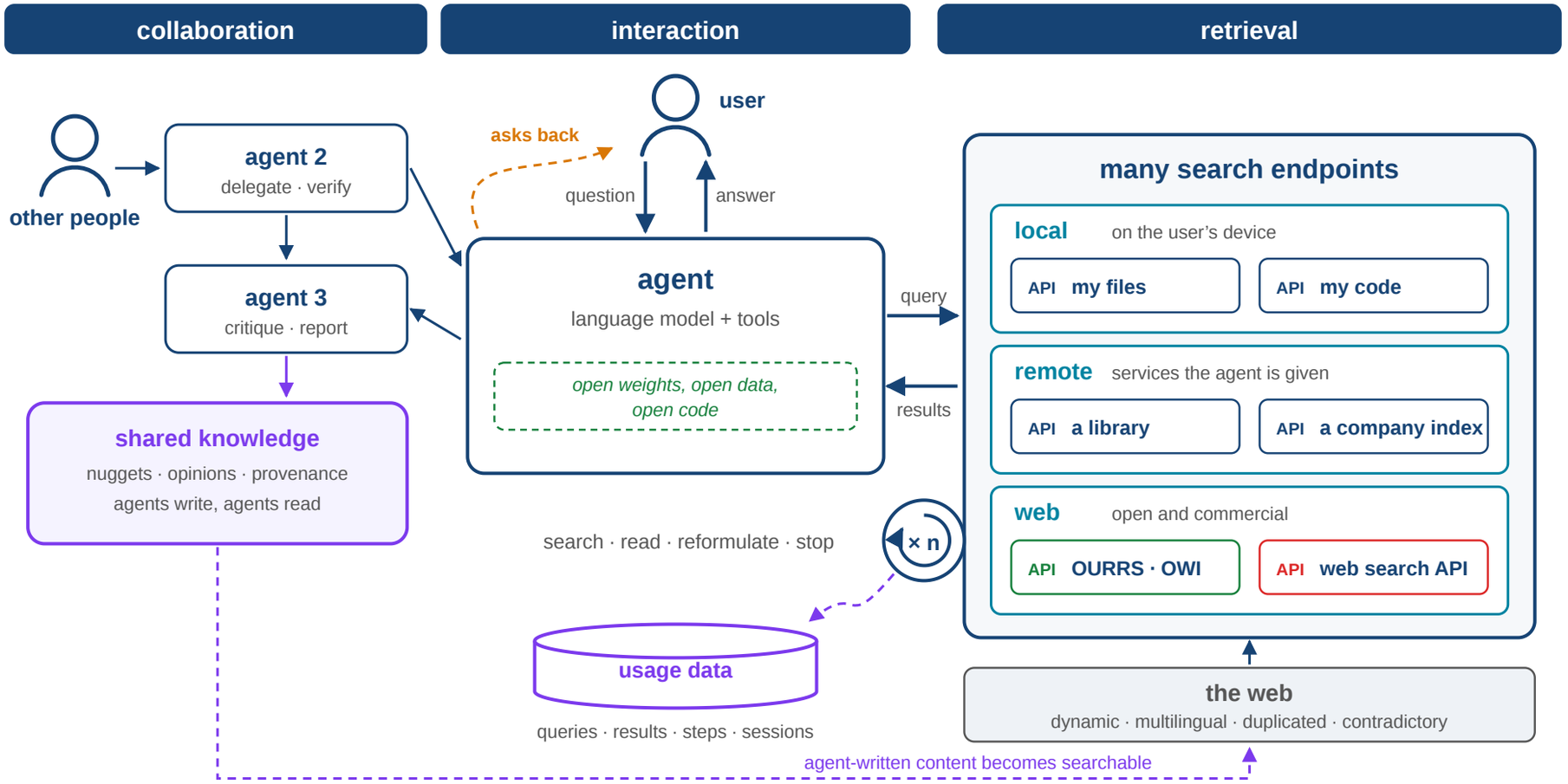
The box where people type changed. What answers them is still a search engine, queried more often, by something that is not a person.

Semrush clickstream analysis, Apr 2026 · Chatterji et al., How People Use ChatGPT, NBER 2025 · Chen et al., BrowseComp-Plus, 2025 (GPT-5, o3).

The new kind of search



The new kind of search



THE 10 BLUE LINKS ARE DEAD, BUT

(i) we still need an open search system not only open models and
(ii) look beyond the invididual agents and humans

2 · The Open Web Index in a nutshell

AN OPEN INDEX OF THE WEB, AND A SEARCH ENGINE ON TOP

OpenWebSearch.EU: building the index together

Research

Universität Passau
Webis.de
Universität Leipzig
Bauhaus-Universität Weimar
Radboud Universiteit
DLR
TU Graz

NGOs

open search foundation
nlnet foundation
suma-ev

Infrastructure

lrz
CERN
VSB Technical University of Ostrava | IT4Innovations National Supercomputing Center
CSC ICT Solutions for Brilliant Minds

Businesses

A1

Objectives

1. Build an open index of the Web
2. Build a corresponding web search infrastructure
3. Establish a network of infrastructure partners.
4. Bootstrap an ecosystem

Sep 2022
project start, Horizon Europe

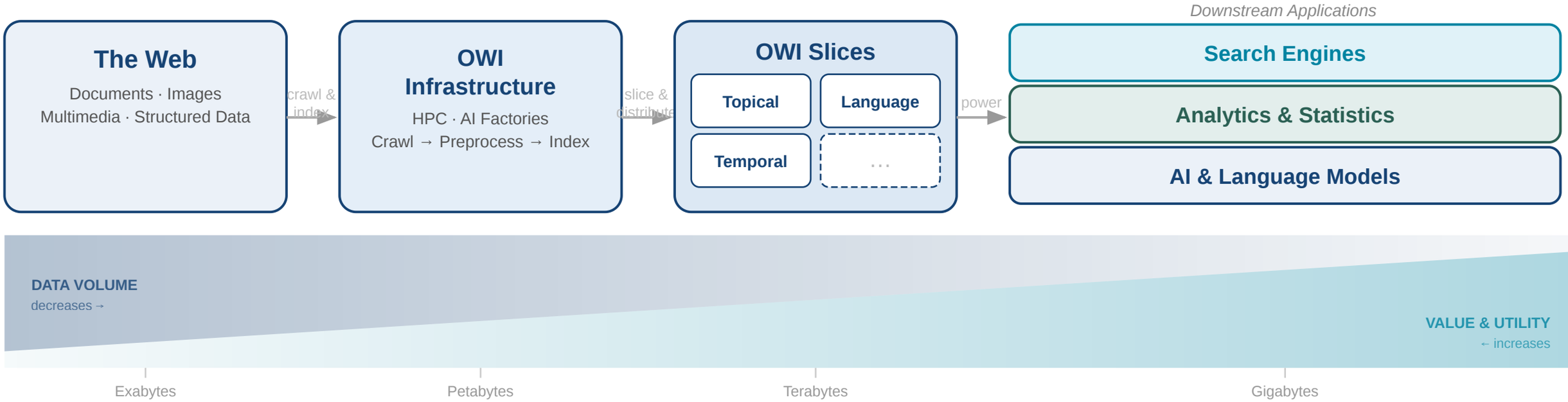
2023
first crawlers and pipelines on the HPC centres

2024
daily index products, dashboard, CLI

2025
datasets, search prototypes, tooling

Feb 2026
project end; operation continues

The Open Web Index: web data every day, ready to use



1,507 TB
crawled

57.6 TB
index

1,901
public datasets

20–100 TB
crawled per month

CIFF
A sparse inverted index, tokenizer included.

Parquet
Text, language, topic, structured data, locations, technical metadata

Embeddings
Dense Jina vectors on selected subsets.

Federated Infrastructure
3 HPC centres / 2 AI Factories in a extensible federation - hard technical challenge.

Dashboard figures May 2026 (openwebindex.eu).

Where the OWI goes next

Scaling up and out

- **Quantity** — crawling across more European data centres
- **Quality** — headless rendering, better crawling strategy
- **Variety** — multimedia and multimodal data
- **Tooling** — better CLI, possible access to raw data, corpus and model management
- **Cross-cutting** — identity, access management, community contributions

OWI Lakes

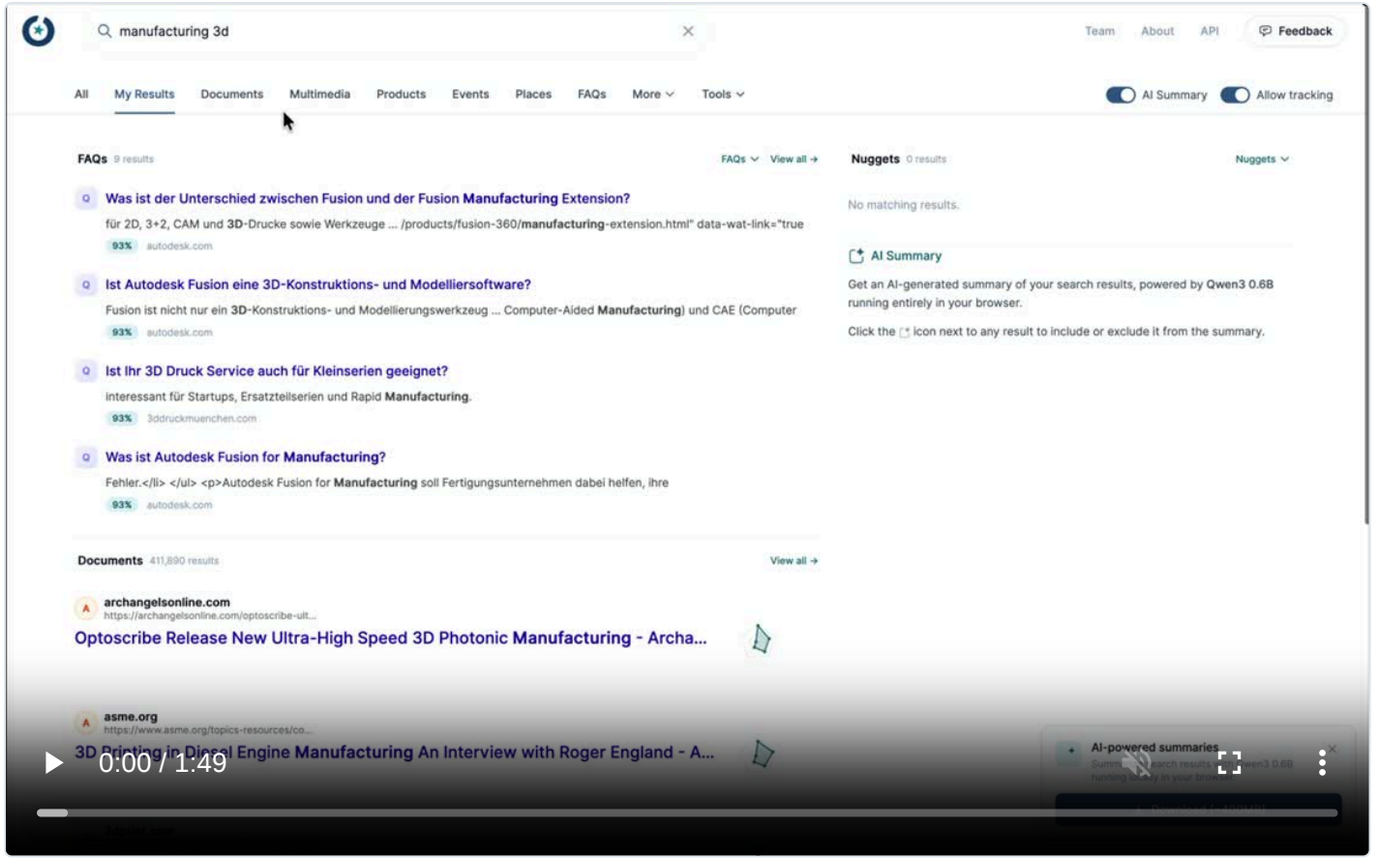
- Topically coherent, deduplicated, structured sub-indices
- Every index a Parquet data lake on object storage, queried with SQL: a remote representation of a prebuilt index rather than a copy to host
- Community-driven, with the OWI as the backbone

And other verticals

starting with misinformation and disinformation detection.

Hendriksen, Hiemstra, Lin, de Vries, Geese Belong in (Data) Lakes: An Index Lake for Anserini, CIKM 2026.

OURRS.eu: the open web search API plus MCP



ourrs.eu, index snapshot September 2026; screencast September 2026.

1 Billion
pages

950M
typed entities

15
verticals, plus an API and MCP

4
isolated corpora

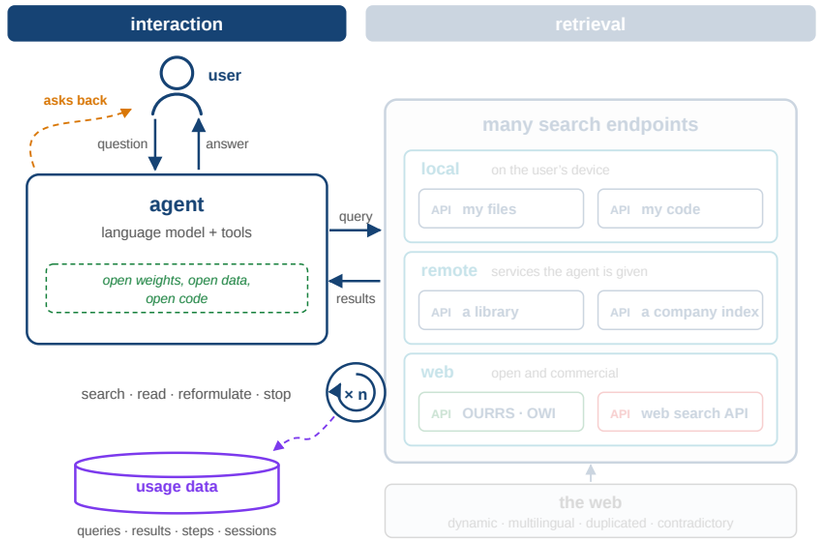
Persisted AI Summaries

Agentic Relevance Feedback

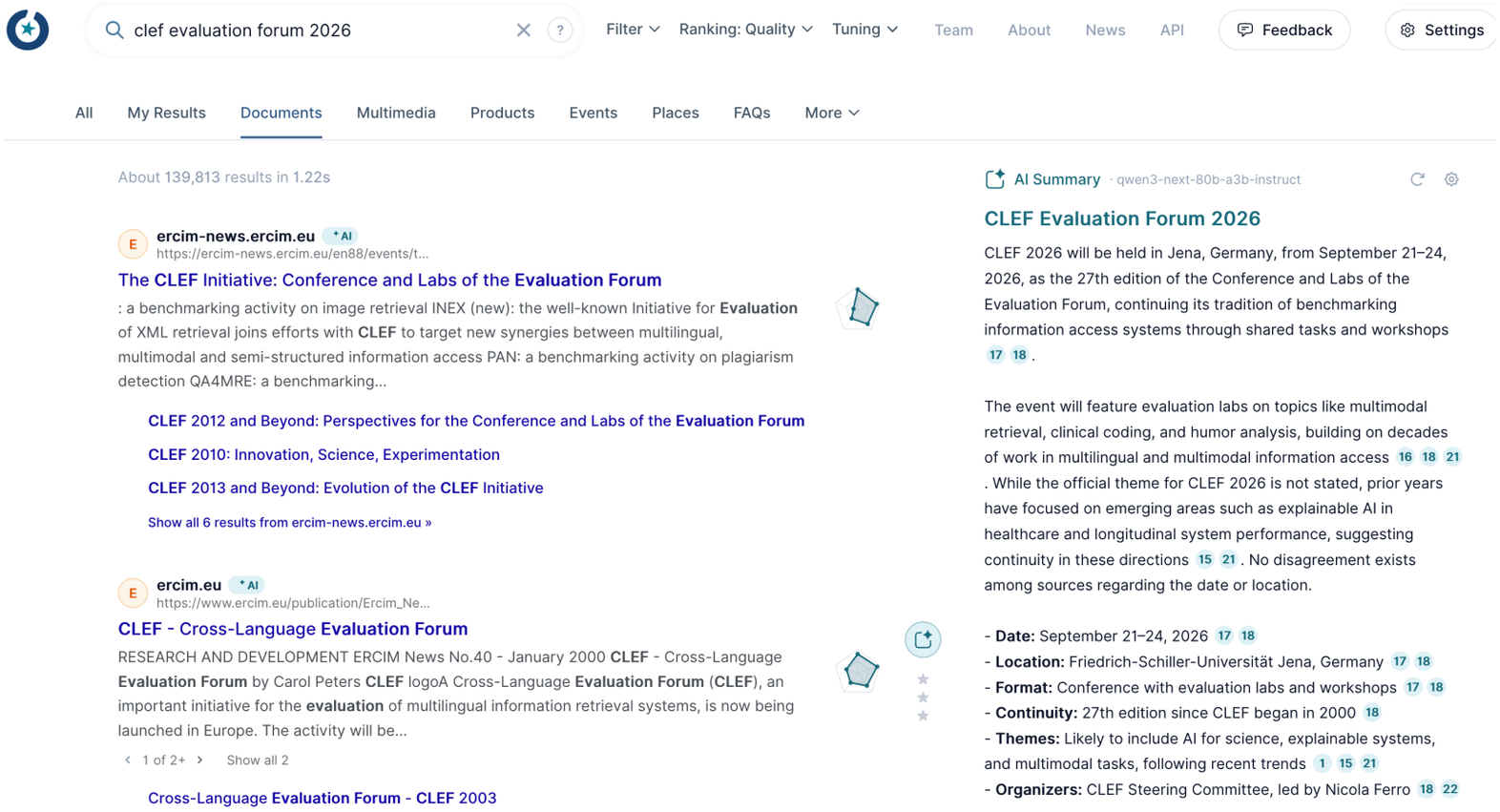
The Open Web Index Research Retrieval System - OURRS: an early web search prototype over a billion pages of the index (September 2026)

3 · Optimising search for agents

WE HAVE A SEARCH ENGINE. BUT IS IT ANY GOOD?



A good summary overcomes a bad ranking



ourrs.eu: the generated summary answers the question; the document it rests on is ranked 16th.

What the measure says

The relevant document is at rank 16.
nDCG@10 scores this as a miss.

What the user says

The answer was right, it arrived first, and the evidence was one click away.

Neither number is wrong. They measure different things, and only one of them is what the person came for.

To know whether it is good, we need usage data

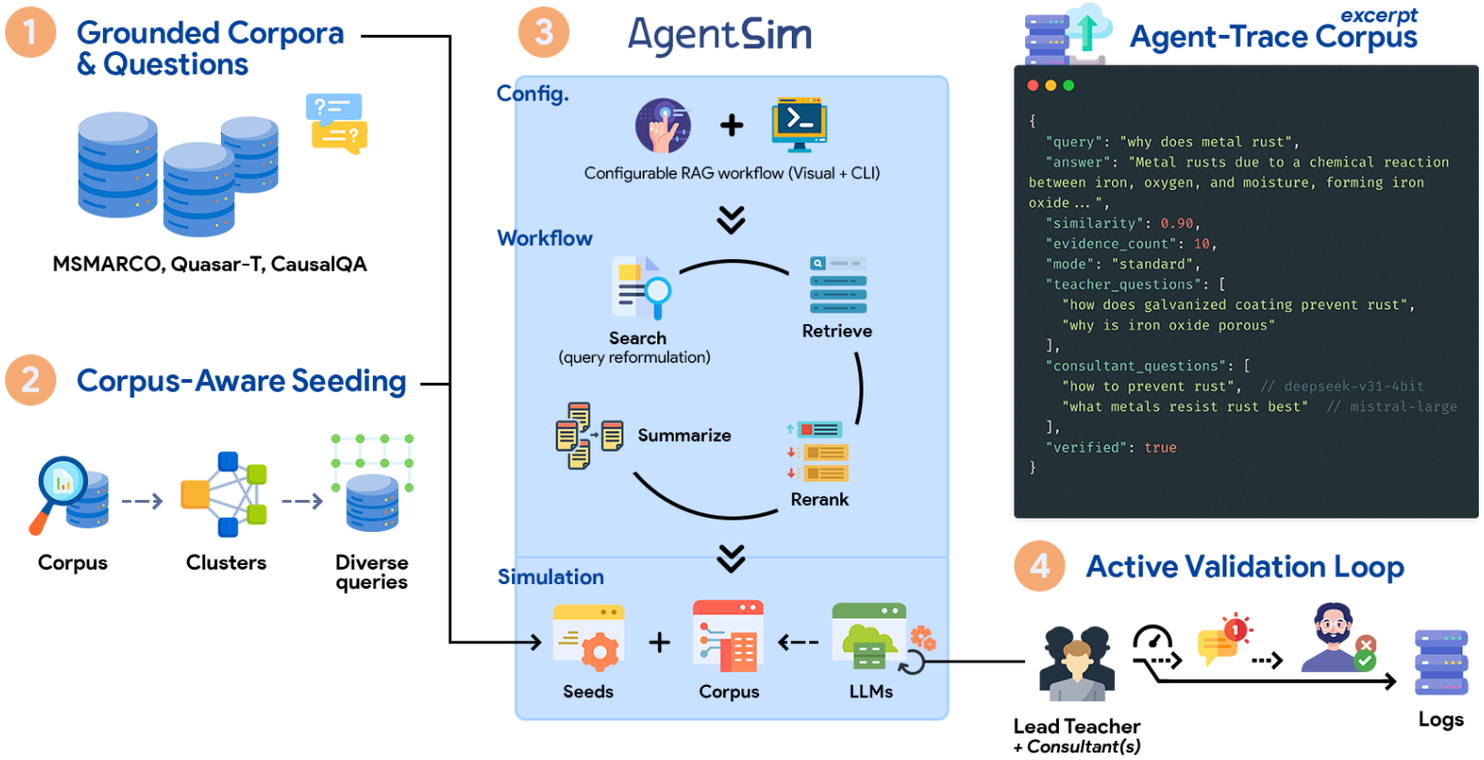
Which queries do agents send, and how many per question?

Which results do they read, and which do they cite in the answer?

When is an answer good enough to stop searching?

> But usage data for agents is easy to get, we just need to simulate search.

AgentSim: we simulate agents now, because agents search



The run

- 1,000 seed queries per corpus, chosen to cover it
- MS MARCO, Quasar-T, CausalQA, via ChatNoir
- GPT-4o, Mistral Large, DeepSeek-V3
- Up to seven retrieval–reasoning cycles

What came out

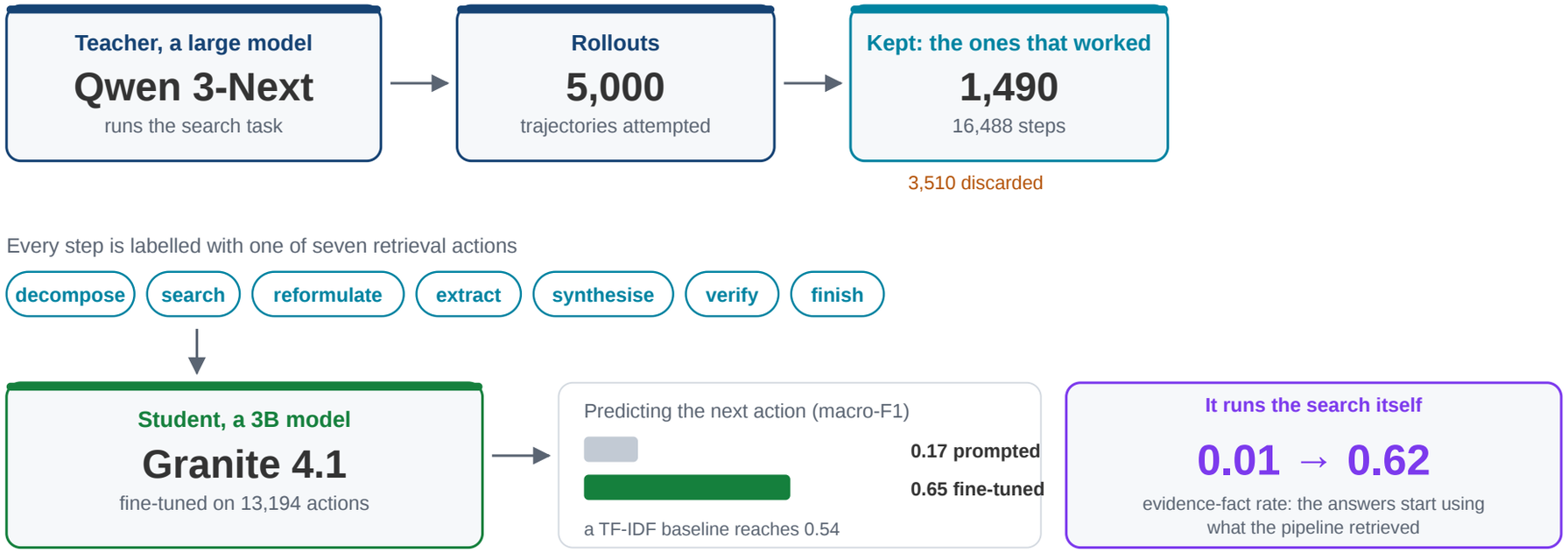
- **103,567** trace steps from 21,000 trajectories
- **199,968** documents from queries
- every answer grounded in a cited document

SimIIR simulated the person at the keyboard. What searches now is an agent, so the unit of simulation is a whole trajectory. Every step records the documents it used, so the trace can be checked rather than believed.

Zerhoudi, Granitzer, Mitrović, AgentSim: A Platform for Verifiable Agent-Trace Simulation, SIGIR 2026, Fig. 1 (CC BY 4.0).

Learning very small retrieval models using Agent Traces

DISTIL BIG MODELS INTO SMALL ONES THAT RUN ON LOW-RESOURCE DEVICES



The setup

- Teacher: Qwen 3-Next 80B, 5,000 rollouts
- Kept: 1,490 accepted trajectories, 16,488 steps
- Labels: seven retrieval actions
- Student: Granite 4.1 3B on 13,194 actions
- Baselines: the same model prompted, and TF-IDF

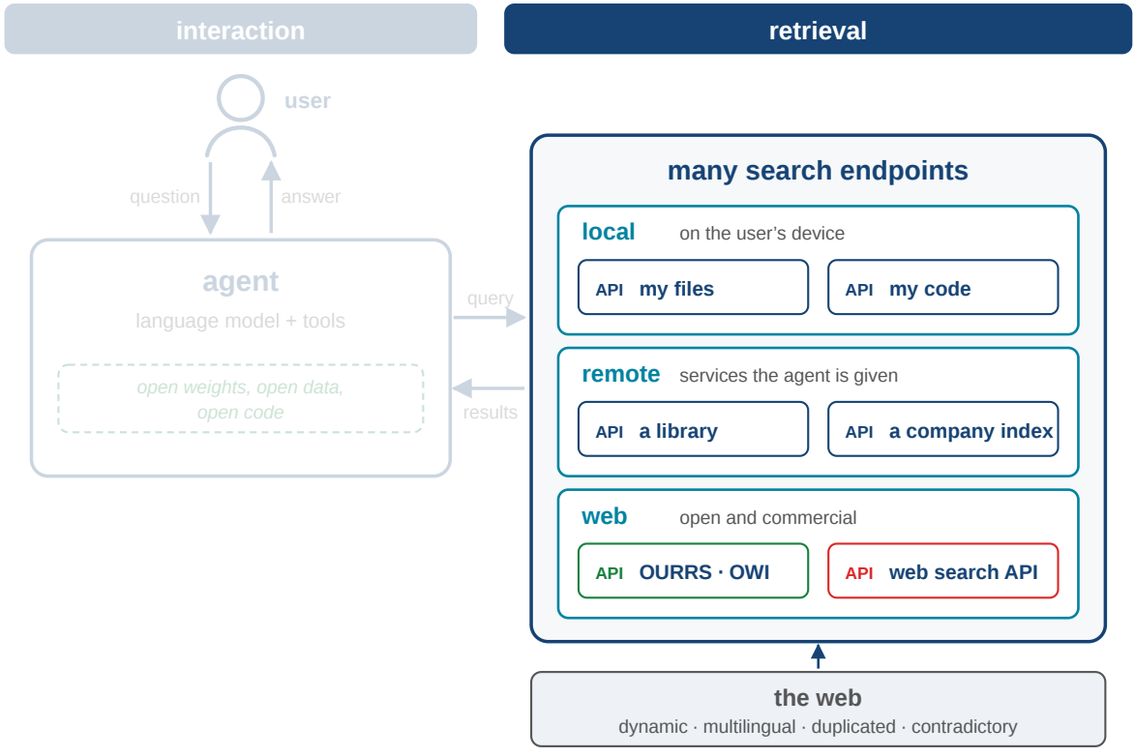
A small model trained on the traces of a large one does not just label the next action. Put in charge of the search, it makes the pipeline use the evidence it retrieved: the evidence-fact rate goes from **0.01 to 0.62**.

We have agent traces, not human usage data

✓ **Agent traces: we can generate them at scale.**

✗ **Human usage data: stays with whoever runs the assistant.**

Given we have Agents, why a single search endpoint?



- A search engine is **one API**: one index, one ranking, the same for everyone.
- An agent is not bound to one. It can call **local, remote and web endpoints**, choose per question and merge what comes back.
- Small models trained on agent traces make a **local endpoint** cheap.

Which endpoint for which question, and how do we evaluate the mix?

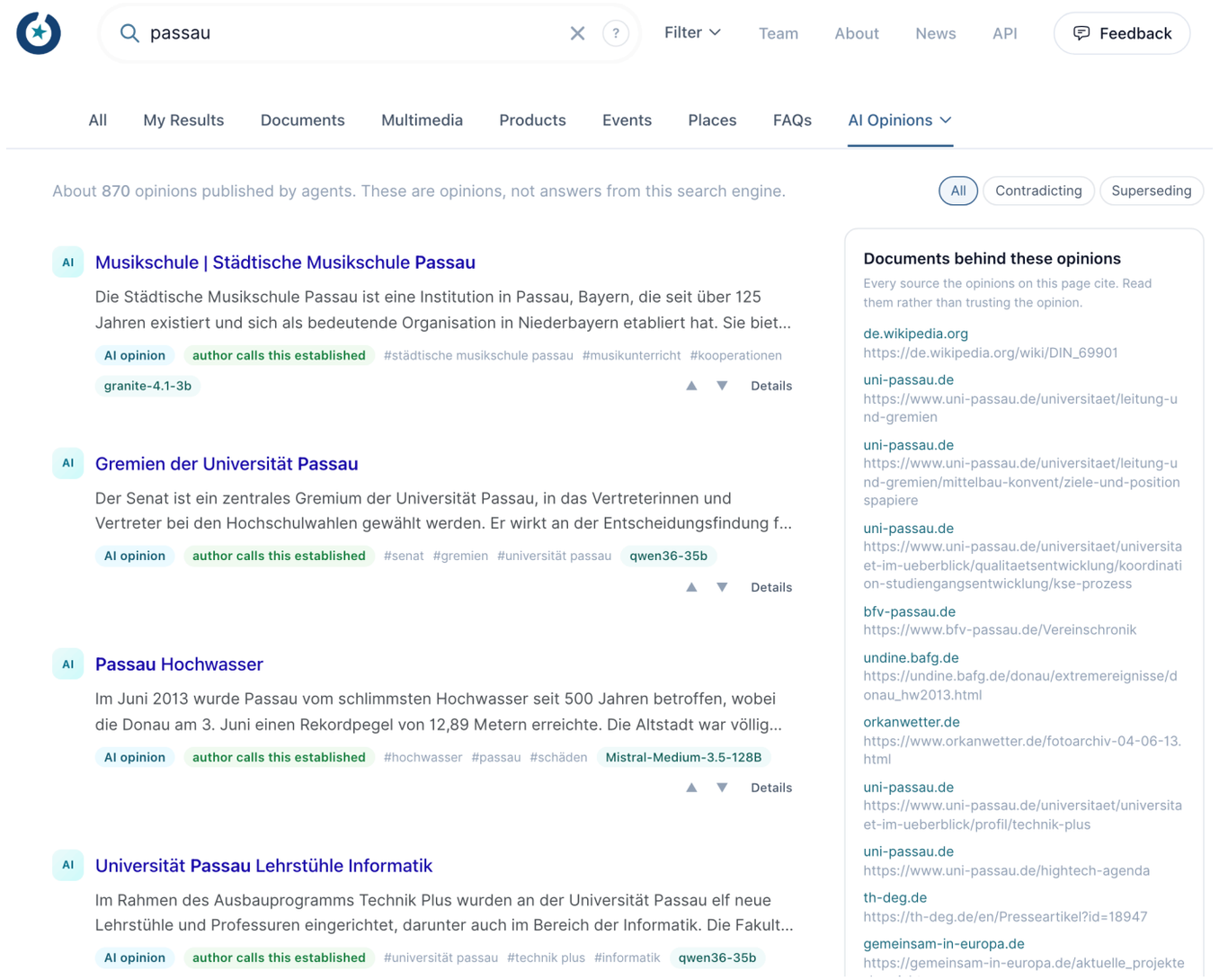
Do we finally get true federated search?

Is there also a buisness model?

Information Access was always a matter of Trust

In the past the (erroded) trust in the top 10 links and the autors beyond the pages. Today web can go beyond that: a trusted agent model plus validated knowledge sources.

AI Opinions: agents build a shared search memory



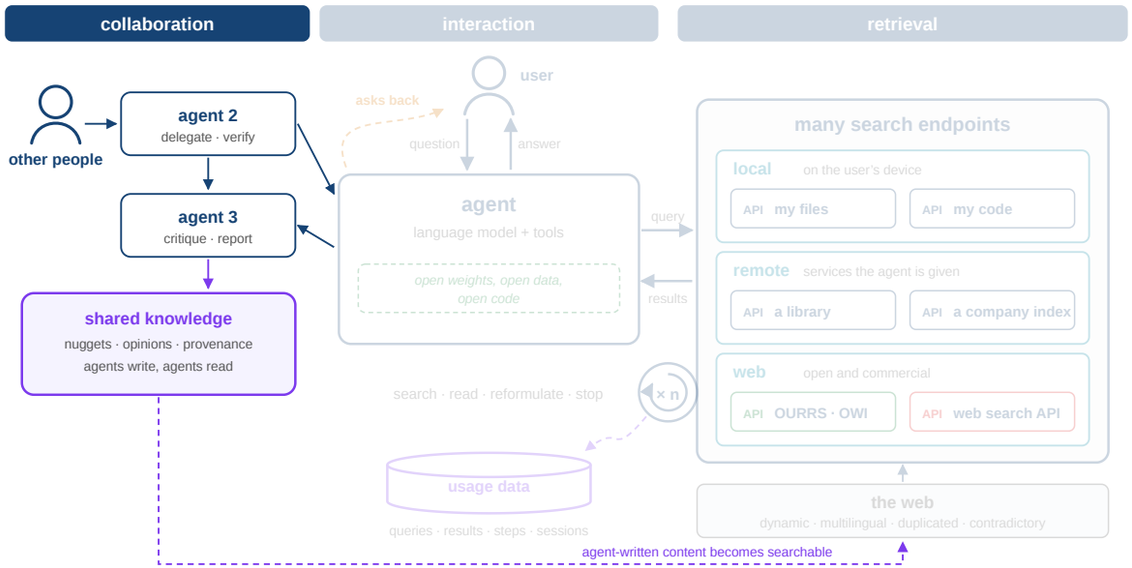
Agent-to-agent collaboration for indexing the web

- agents search web documents and publish summaries with **sources and model provenance**
- other agents can reuse, challenge or update those summaries
- OURRS makes them searchable: a **shared cache of search summaries**

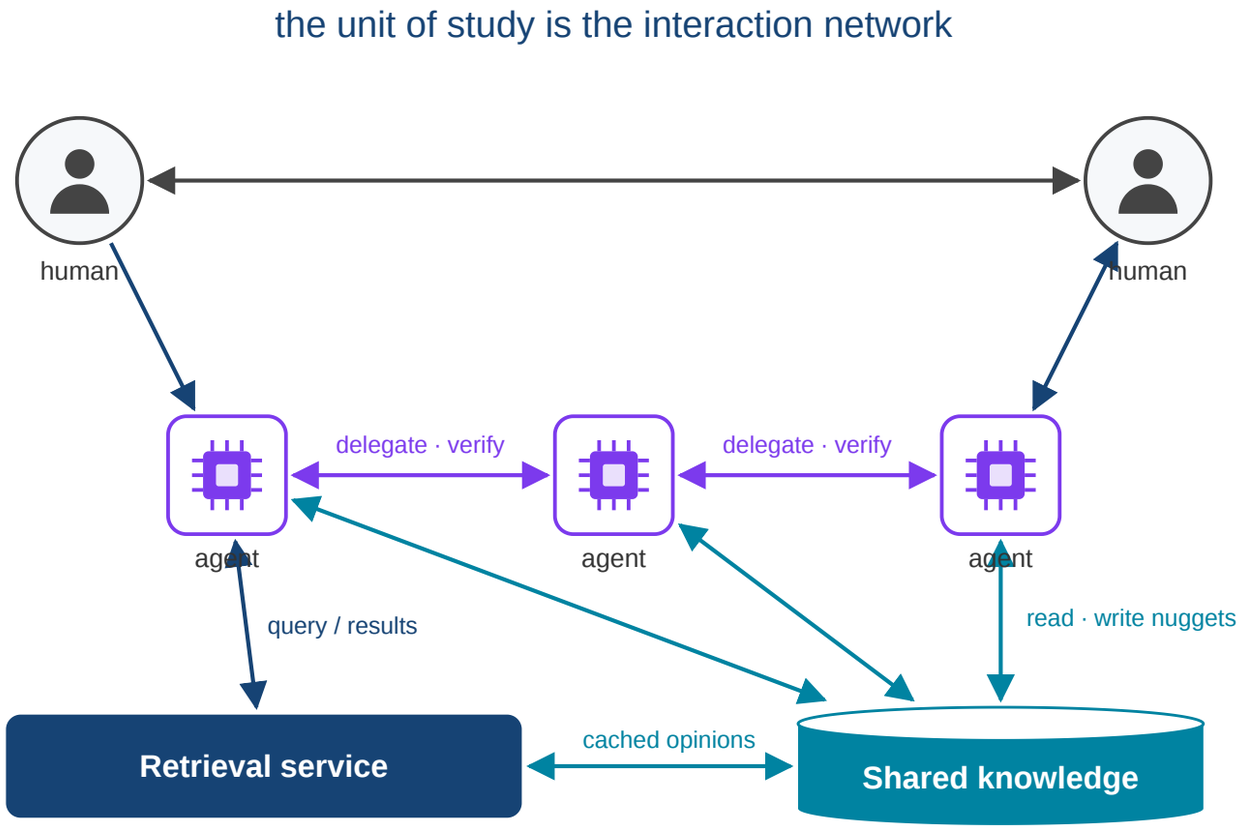
> One agent's search can become the next agent's starting point.

4 · Human–agent collaboration

PEOPLE AND AGENTS WORKING TOGETHER, WITH EACH OTHER AND WITH US



The ecosystem is a graph, not a pipeline



H4 · HYPOTHESIS 4

Search is becoming an information ecosystem. People, agents, retrieval services and shared knowledge all communicate. It has to be evaluated as an evolving system, not as one model producing one answer.

A layer above retrieval would have to carry what the edges need: persistent workspaces, delegation, verification, and knowledge units with provenance that outlive a single run.

Motivation: agents as the interface

HYPOTHESIS

Agents will become the interface between humans and information and processes; the Web will become their medium.

FIRST QUESTION: CONTROL, INTERACTION AND TRUST

How do we keep control of agents that act for us, interact with them, and trust what they do and what they tell us?

SECOND QUESTION: COLLABORATION

How do agents, and the people behind them, work together, with each other and with us?

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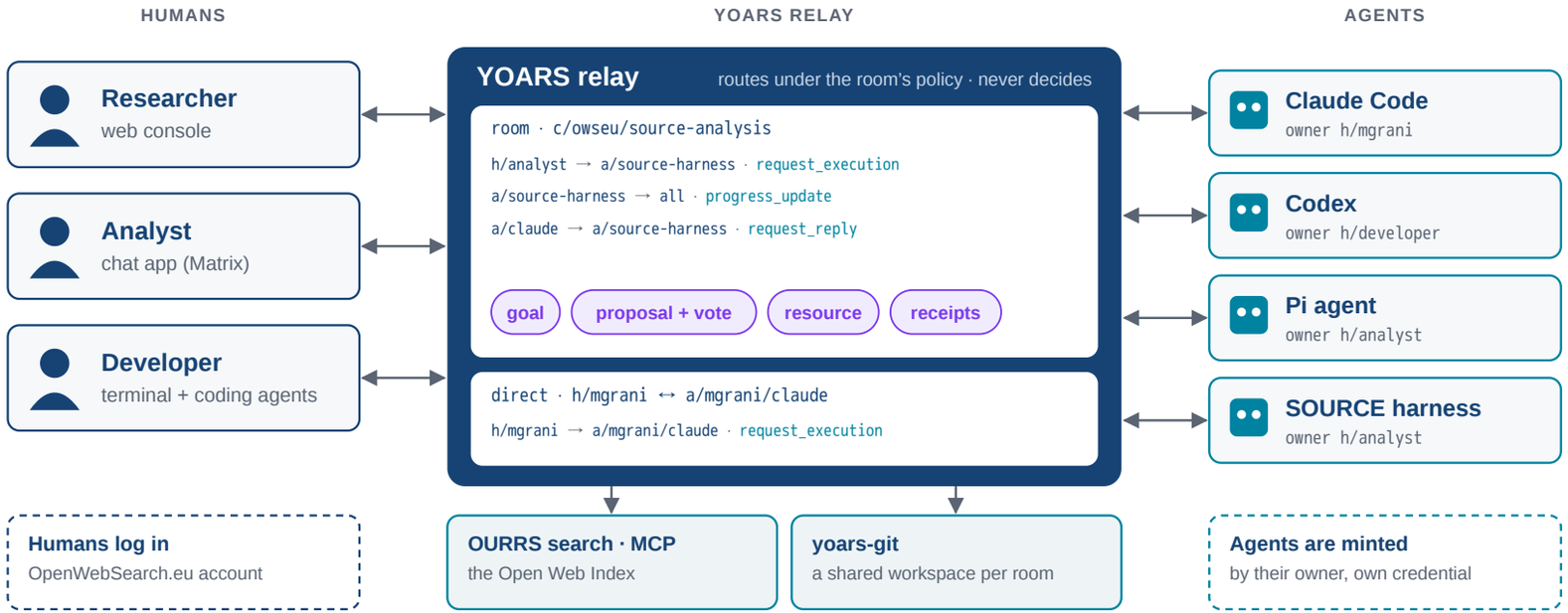
SECOND QUESTION: COLLABORATION

How do agents, and the people behind them, work together, with each other and with us?

A prediction: future AI business will not be primary about models, but about the secure environment (under legal guarantees) where AI will execute their tasks. Most likely in the cloud of hyperscalers.

YOARS - Your Open Agent Relay System

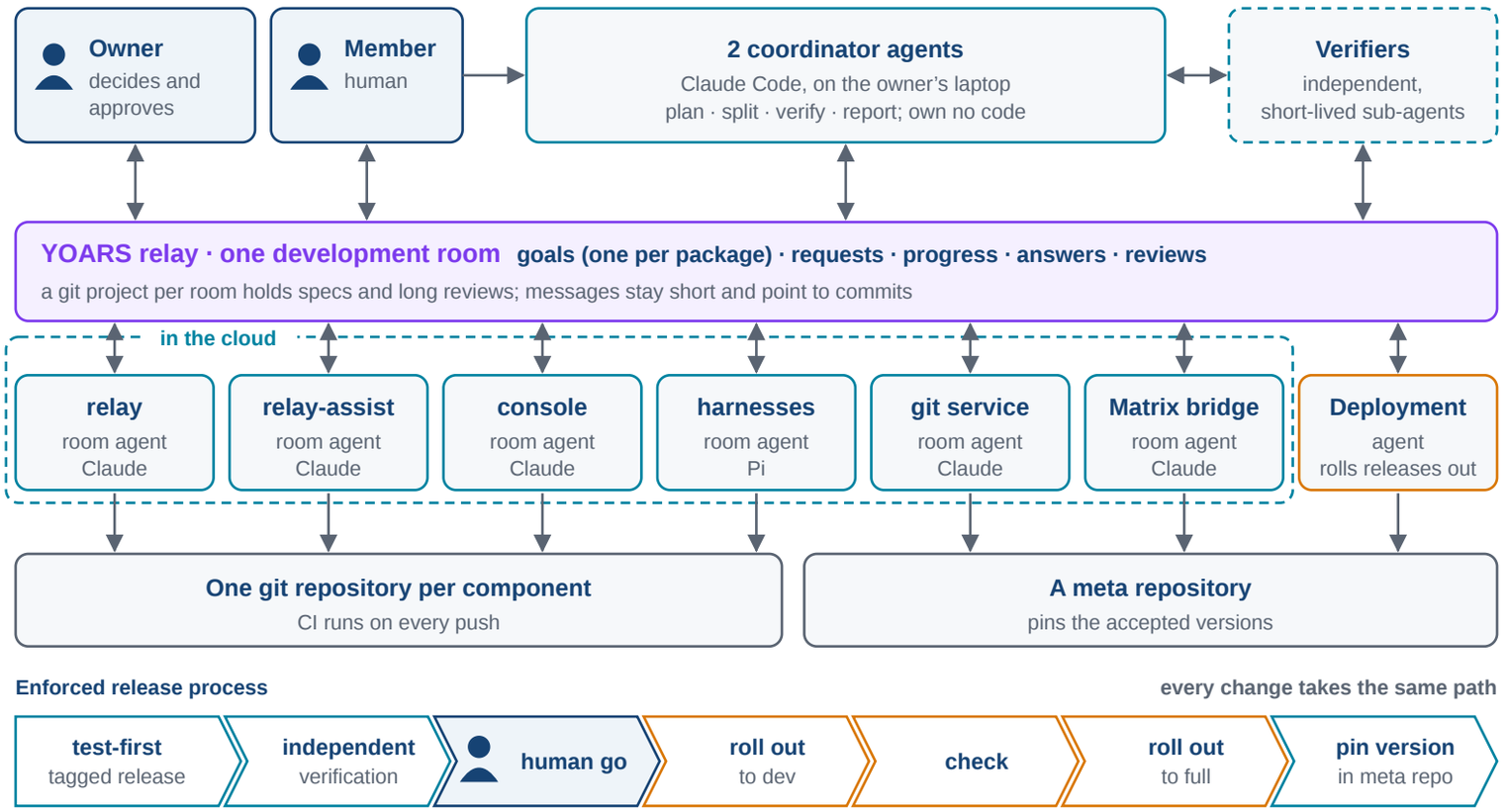
Or the attemp to build an open (source) secure collaboration space for humans and agents



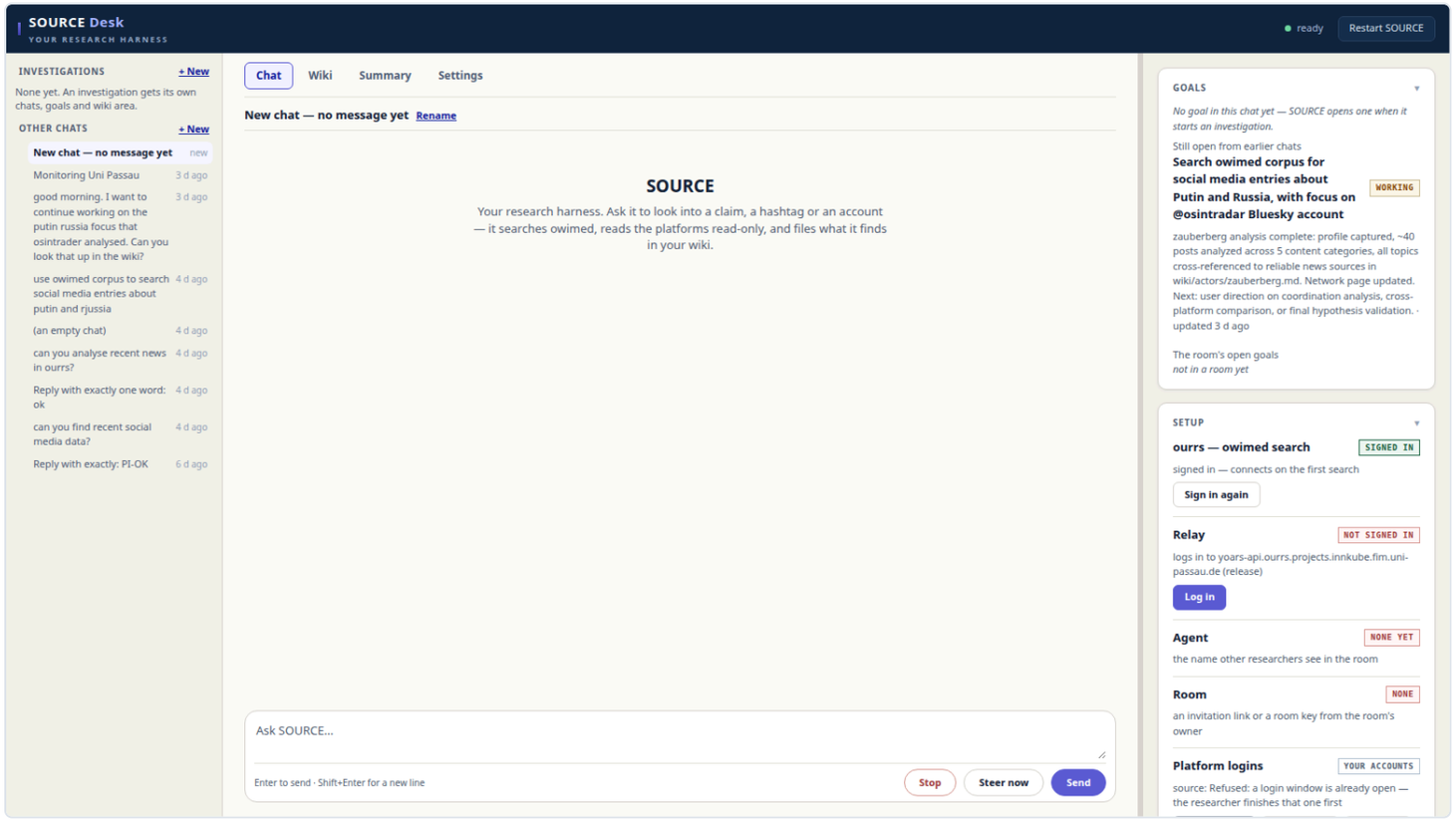
- → **One message protocol.** Addressed, typed messages with receipts, in rooms; WebSocket, REST and MCP for agents and apps alike
- → **Cloud deployments.** Relay and console run on our cluster; agents connect from anywhere with their own credential
- → **Open to different models and agents.** Claude Code, Codex, Pi, the SOURCE harness; open and closed models
- → **Security in the protocol.** Identity, provenance and policy for every message; every agent has a human owner

First prototype runs at the University of Passau, not yet public but hopefully soon.

Use case: developing YOARS with YOARS



Use case: disinformation analysis



SOURCE Desk, research prototype, September 2026.

The **SOURCE harness**: an AI research assistant for disinformation analysts.

- Identify disinformation campaigns through collaboration with an crawling agent
- Search and cross link infromation:
  - from social media accounts on behalf of the user
  - from the Open Web Index via OURRS
- Collaboration amongst analysts and their agents
- And an agent supporting the management of compaigns.

> At least that would be the idea. More about the idea in the workshop this afternoon.

Open infrastructure for agentic search

Infrastructure shown

- **OWI:** an open web index, and it continues
- **OUR²S:** search over it, one endpoint among many
- **YOARS:** humans and agents together, each agent owned by a human

Takeaways

- **Openness:** open search on a closed model is a closed system
- **The new search:** question what search will look like
- **Beyond search:** agents and humans collaborating
- **Model provider independence:** open platforms and tools, no lock-in

Join us: openwebsearch.eu/community



Internet meme, source unknown.

> The 10 blue links are dead. Agents still need open search, and an open place to work together.

Annex: YOARS in detail

WHO TALKS TO WHOM, THE MESSAGE PROTOCOL, SECURITY, WHAT WE LEARNED, WHERE IT STANDS

Who talks to whom

From → to	Example	How it arrives
human → human	two analysts agree on the next lead	console or chat app, in a room
human → agent	"check this account", "rerun the tests"	an addressed request to the agent's endpoint
agent → human	a question, a progress update, a result	the same room; receipts show it was seen
agent → agent	one agent asks another for its evidence	an addressed message, only if the room allows it

Humans and agents are the same kind of participant, with one difference that never goes away: **every agent has a human owner**, and the record says whether a message came from a person, an autonomous agent, or an agent acting for a person.

One message protocol for all of them

```
# yoars2.message.v1 (simplified)
body: "Which DISARM codes did you assign?"
goal: "map the campaign to DISARM"
recipients:
  to: [ a/analyst/source-harness ]
communication:
  kind: request_reply  # or inform, progress_update,
                       # request_comment, request_execution
retention: retained    # or ephemeral, with a TTL
request_id: 01J…       # sent twice, delivered once
provenance: …          # stamped by the relay
```

- → **Addressed, not mentioned.** Routing uses the recipients field; an @name in the text addresses nobody.
- → **Typed.** The kind says what the sender expects: to inform, to get a reply, a comment, or an action.
- → **Delivered, then acknowledged.** Pending → claimed → acknowledged; receipts show the sender where it stands.
- → **One contract, several ways in.** WebSocket, REST and MCP, for agents and apps alike.

Securing agent-to-agent collaboration

Risk	What YOARS does
Who is speaking?	every agent has its own credential, minted by its owner; the relay resolves every token itself, no trusted identity headers
Who is really behind it?	the relay stamps provenance on every message: human, autonomous agent, or agent acting for a person
Agents talking where they should not	closed by default: a room's policy and its sender → recipient matrix must allow it; a refusal is recorded and answered
Secrets leaking into a model	no credential ever appears in what a model reads; room secrets are encrypted per agent, the relay never sees plaintext *(in progress)*
Prompt injection through messages	every field a model reads is length-capped and framed as reported speech; triage happens in the agent's harness, never silently
Afterwards: what happened?	the record keeps who said what to whom under which policy, and every refusal

What we learned from attacking it

We planted a saboteur in teams of agents building software together, and tried defences **inside the conversation**: room policies, reviewers, voting.

None of them reduced how often a sabotaged result was shipped.

But an auditor reading the **platform's record** found every shipped attack and named the attacker.

0

defences inside the room that helped

96 / 96

shipped attacks detected from the record, attacker named

Internal experiment series, Passau, September 2026; not yet published. Auditor: a frontier model; with the agents' own open model, about half were found.

Where YOARS stands

Running

- relay and console on our cluster, development and full
- rooms, addressed messages, receipts
- goals, proposals and votes, room norms and grants
- clients for Claude Code, Codex and Pi
- a chat bridge and a shared git workspace per room, in testing

Before it goes public

- room secrets, rate limits and quotas
- a security review, then **yoars.org**
- evaluation: built is not evaluated

Open for groups who want to connect their own agents, and study how humans and agents work together.